

SPECIAL ISSUE PAPER OPEN ACCESS

Predicting Arbitrage Occurrences With Machine Learning and Improved Decision Threshold Level in Live-Trading Crypto Environments

Kristína Okasová | Michal Géci | Kristián Košťál 

Faculty of Informatics and Information Technologies, Slovak University of Technology in Bratislava, Bratislava, Slovakia

Correspondence: Kristián Košťál (kristian.kostal@stuba.sk)

Received: 1 September 2024 | **Revised:** 27 May 2025 | **Accepted:** 29 September 2025

Keywords: AI | arbitrage | blockchain | cryptocurrencies | exchanges | peer-to-peer networks | trading

ABSTRACT

Cryptocurrencies represent a significantly utilized class of digital assets, encompassing a diverse array of tokens and coins available for trading purposes. In this study, the integration of machine learning algorithms with an arbitrage trading strategy across cryptocurrency exchanges is explored. The objective is to scrutinize prominent cryptocurrency pairs characterized by high volatility, vulnerability to speculation, regulatory gaps, liquidity constraints, and heavy-tail distribution, with the intention of training the model to predict the potential for arbitrage. To differentiate from competitors who await rare arbitrage opportunities, a novel approach is introduced to enhance arbitrage profitability. The primary innovation of this study lies in demonstrating the capability to predict profitable arbitrage opportunities in discrete intervals in advance, by incorporating sophisticated confidence level metrics to initiate arbitrage trades only when the model's predictions demonstrate substantial certainty. The findings indicate that the profitability of the entire strategy exceeds 100% within a 1-week timeframe.

1 | Introduction

With the increasing popularity of cryptocurrency trading, diverse methods and strategies are emerging. Arbitrage is among these, focusing on profiting from price differences across various exchanges or cryptocurrency pairs. Although this trading method is not new, it occurs infrequently with fiat currencies. However, given the relatively young cryptocurrency market and the multitude of pairs available, arbitrage opportunities can arise more frequently, offering a higher chance of profit [1]. To guarantee profit capture, the entire operation must be carried out concurrently and repeatedly [2], rendering this strategy impractical for human traders and necessitating the implementation of an arbitrage bot. This automated trading method retrieves data from a selected cryptocurrency exchange through its application programming interface (API) and available endpoints, providing the needed data in JavaScript Object Notation (JSON) or Extensible Markup Language (XML) formats [3].

To elucidate the arbitrage process in more detail, it involves the continual buying or selling of cryptocurrency assets on one exchange or within a trading pair while simultaneously executing the reverse transaction on another exchange before the profit margin dissipates [4,5]. The competitive nature introduced by numerous arbitrage bots highlights the advantage of applying machine learning (ML) models to forecast profitable opportunities in advance. Cryptocurrency prediction could potentially benefit from employing the following algorithms [6]:

- Logistic regression—a classification model that evaluates the relationship between a binary result and a dataset of any number of independent variables [7],
- Random forest—an algorithm applicable to both classification and regression tasks, relying on the average or the most frequent result from the collection of decision trees that constitute the random forest [8, 9],

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2025 The Author(s). *International Journal of Network Management* published by John Wiley & Sons Ltd.

- Support vector machine (SVM)—a classification model that attempts to find an optimal boundary to distinguish observations of different classes [10],
- Multilayer perceptron (MLP)—known as one of the simplest forms of neural networks (NNs)—contains layers of neurons through which information travels from input through hidden layers to the output layer, with weights that need to be learned [11].

Different approaches to integrating the specified ML algorithms were explored to determine the most lucrative by varying the time intervals of the collected data and adjusting the hyperparameters for each algorithm individually. The key benefit of this implementation lies in immediate transactions, aiming to secure potential profits before they are lost. This could be the primary advantage of this paper, as it accelerates the arbitrage search and leads to more exploited opportunities compared with other trading bots. When ML algorithms can forecast an exploitable arbitrage opportunity even before it materializes, the purchase can be executed at the start of the analyzed period, ensuring it is among the first to maximize the profit.

This paper is an extended version of our previous work [12]. We extended the last paper by a new threshold level, new datasets, a decision-making process, applying more sophisticated risk management, and new evaluation and comparison to increase the possible profits from using it.

2 | Background

Arbitrage involves buying an asset in one market while simultaneously selling it in another at a different price [4]. Such price differences are common in highly volatile, unstable, decentralized markets with minimal regulations. These characteristics apply to cryptocurrency markets, bolstered by the rise of numerous currencies and exchanges. This scenario facilitates trading and allows the entire process to be algorithmized, exploiting limited arbitrage opportunities. To mitigate the risk associated with this strategy, small transactions are carried out more frequently, yielding lower but more consistent profits [1, 2, 13]. The factors contributing to arbitrage opportunities can be described as follows.

- Market efficiency,¹ which is developing gradually, still offers numerous profitable arbitrage opportunities [14],
- Differences in prices among exchanges lead to deviations from the volume-weighted average price [14],
- The currency spread, or the difference between the bid and ask price, expresses the contrast in willingness to buy or sell a specific asset and, together with significant order imbalance, leads to higher spread [13, 15],
- Specific patterns, considering the impact of the time of the day, liquidity, exchange heists, and hacks [14].

On the other hand, several factors eliminate arbitrage opportunities. High trading fees can consume profits, moreover, the significantly stronger position of the buying side over the selling one often narrows the spread for arbitrage. Additionally,

predicting arbitrage opportunities is complex due to the lack of correlation between their size, frequency, and the number of exchanges involved [13].

No universal method exists for identifying arbitrage opportunities, as various factors influence the process. Trading on exchanges, particularly with diverse assets, may offer advantages [16]. While market crashes can occasionally create arbitrage opportunities, they are rare [17]. Leveraging blockchain's decentralization enhances the potential for discovering and exploiting arbitrage opportunities [18].

2.1 | Cryptocurrency Market

Cryptocurrency markets' unregulated, decentralized nature presents significant drawbacks for end-users, including high uncertainty, speculative components, volatility, liquidity issues, and technical challenges [19–21]. Over time, these markets have matured, with reduced volatility, increased trading volumes, and more outstanding investor commitment [21, 22]. This progress is driven by decentralization, transparency, limited regulation, and open access to innovation [23, 24].

While cryptocurrency spot markets dominate trading activity, futures markets have recently emerged. Although their volume remains limited compared with spot markets, inflation expectations and the ability to manage risks between hedgers and speculators may boost their popularity. Futures markets demand sophisticated strategies involving active rebalancing and margin management, offering a hedge against spot market volatility [25, 26].

Perpetual futures, a key derivative in exchanges with significant cryptomarket share, combine the continuous availability of trading with cryptocurrency decentralization. Denominated in stablecoins [27], they offer high leverage and payouts, attracting traders. These markets efficiently maintain daily price equilibrium through funding rates, penalizing dominant positions, and deriving value from short-term trends. However, these features can also be exploited by market manipulators trading against low-volume noise traders [24, 26].

The inherent volatility of cryptocurrencies and market inefficiencies in futures markets create arbitrage opportunities. This study focuses on utilizing perpetual futures to capitalize on arbitrage opportunities, gradually diminishing as cryptocurrency markets mature [26].

2.2 | Market Efficiency

Market efficiency varies across assets and has evolved significantly in the cryptocurrency market. Before 2017, it was highly inefficient, but as markets matured and traders gained experience, efficiency has steadily improved [22, 28]. Efficiency also fluctuates with crises, bitcoin's (BTC) evolution, and economic policy uncertainty [23, 28].

Several factors influence this evolution. Global financial stress negatively impacts efficiency, while liquidity and money flow

positively affect it, with the latter being relevant only in already efficient markets. Other factors include volatility, order flow, and investment shifts to substitutes like gold or energy. Uncertainty also plays a critical role in shaping market dynamics [13, 28, 29].

Market efficiency directly affects asset attractiveness. Highly efficient cryptocurrencies are perceived as more reliable and reputable, while inefficiencies indicate poor information quality and lack of regulation, potentially leading to a herding effect [23, 28].

2.3 | Cryptocurrency Volatility

Understanding cryptocurrency market behavior requires analyzing both efficiency and volatility. Volatility, expressed as price deviations, is central to financial analysis and essential for identifying profitable trading opportunities. Cryptocurrency volatility far exceeds fiat currencies, making accurate estimation and forecasting critical. News and speculation significantly influence volatility, with an asymmetric impact: while equity markets react more to negative news, cryptocurrency markets are more sensitive to positive shocks. This is largely due to uninformed noise traders reacting strongly to positive news, creating herding effects. However, for major cryptocurrencies like BTC and ether (ETH), informed investors increasingly dominate, reducing this effect [19–21, 29, 30].

As cryptocurrency markets mature and align more with traditional financial markets, volatility decreases. Positive news now tends to stabilize volatility, while negative, fear-driven news amplifies it. Despite this trend, cryptocurrency markets retain high volatility persistence, meaning periods of high volatility are likely to last longer [21, 23].

Cryptocurrency trading carries significant risks due to high volatility, susceptibility to speculation, lack of regulation, liquidity issues, and heavy-tail behavior. Cryptocurrencies are better categorized as innovative financial and blockchain-native assets than traditional currencies, which are largely uncorrelated. Their volatility stems from pure market demand without guarantees and their sensitivity to political and economic events [13, 19, 21, 29, 30].

The market's evolution suggests cryptocurrencies are here to stay, though only the strongest will lead. This study focuses on leading cryptocurrencies like BTC and ETH, chosen for their high volatility and transaction volume, using USD Tether (USDT) as the quote asset [31,32].

BTC's prices exhibit the largest changes, reflecting higher volatility than other cryptocurrencies. These fluctuations are driven more by transaction volume than speculation [21, 29, 31]. Persistent price deviations between exchanges create opportunities for arbitrage trading [13].

The efficiency of ETH is, similarly to BTC, driven by financial stress, the world stock market, and its volatility [28]. Due to these facts, BTC and ETH also tend to behave slightly differently compared with other cryptocurrencies, mainly due to the asymmetry of their volatility impact, which is likely to increase as negative news appears and, at the same time, copies the behavior of the stock markets [19]. These cryptocurrencies can rely on well-developed infrastructures and established exchanges that cause the highest trading volumes and liquidity levels among all existing digital assets. Another key aspect of choosing these two cryptocurrencies with USDT as their quote asset was the top ranking in the number of profitable arbitrages for the BTCUSDT and ETHUSDT pairs. Moreover, these markets reflect the general mood of cryptocurrency through their behavior and development [13].

3 | Related Work

Various studies are dedicated to arbitrage bot trading on the cryptocurrency market. Only a small number of them are trying to promote the strategy using ML algorithms to predict a profitable arbitrage opportunity even before it occurs. The existing solutions focusing on a similar topic as this paper or its previous version [12] are summarized in Table 1 and described in more detail below. The real-time comparison of their trading abilities with our solution is impossible as some of the solutions are closed-source, and others provide only services for customers interested in incorporating the ML arbitrage bot tailored precisely to their needs.

According to T. G. Fischer et al. [6], implementing logistic regression or random forest could be beneficial. The highlights

TABLE 1 | Comparison of the related work.

Solution name/ author	T. G. Fischer et al.	Madan, I. et al.	WunderTrading	Antier	FreqTrade
ML inclusion	Logistic regression, random forest	Binomial logistic regression, random forest, support vector machine	Machine learning classifier	Any requested	Logistic regression, multilayer perceptron
Commercial solution	No	No	Yes	Yes	Yes
Number of exchanges	6	2	16	23	8
Cryptocurrency pairs	Top 40 coins by market capitalization	Only bitcoin	Any accessible	Any accessible	Any accessible

presented in this paper align with the findings of our research. These include the fact that the most significant impact is determined by the minute-binned price, volume obtained, and the highest market capitalization of the considered cryptocurrencies to reflect the market volatility and meet the minimal liquidation criteria. The choice of ML algorithms is mainly about comparing transparent and black-box approaches. The results have shown that random forest outperforms logistic regression in every metric utilized to measure profitability, identically to the results provided by this paper.

The paper by Madan, I. et al. [33] is dedicated to BTC price predictions and focuses on the relevance of available cryptocurrency features. Again, choosing an ML algorithm is not crucial. However, many different features are collected, mainly considering the number of transactions or blocks, transaction fees, or trading volume, which were not considered essential for our paper. For the price movement prediction, binomial logistic regression, random forest, and SVM were implemented and trained on price data for various time intervals. The outcomes favor binomial logistic regression when working with daily data and random forest for intraday price data collection, which partially agrees with our findings.

In addition to the research publications, multiple commercial solutions are available, mostly combining various algorithmic trading strategies that cover arbitrages. These solutions utilize similar functionalities to our arbitrage bot but provide a much broader scale of trading strategies and ML algorithms. One possibility published by WunderTrading offers multiple trading bots, one incorporating artificial intelligence (AI). It relies on historical and real-time market data to uncover relationships and dependencies, providing insights overlooked by algorithmic approaches and human trades. The key point of the AI analysis is centered on the spread between prices of two comparable assets that benefit from its deviation, similar to the solution proposed in this paper. Moreover, to support the risk-less characteristic of arbitrage trading, the Take Profit and Stop Loss targets are incorporated, resulting in the maximization of profit and minimization of losses simultaneously [34].

The commercial solution by Antier is centered solely around arbitrage trading. The company focuses on developing cryptocurrency arbitrage bots, providing a Bot-as-a-Service model to enable businesses to incorporate trading bots that fit their needs. The AI aspect relies on data analysis and pattern recognition, providing fast execution to take advantage of every arbitrage possibility. Moreover, backtesting of trading strategies, stop loss, and push notifications are available. In addition to concurrency, multiple types of arbitrage are covered, such as triangular, spatial, inter- and intra-exchange, or decentralized arbitrages [35].

Other relevant approaches involve diverse arbitrage bots aiming to manage trading strategies sensitive to transaction timing algorithmically. While these do not incorporate any ML approaches, searching for arbitrage opportunities and calculating potential profits is the same. This is a source of inspiration and validation for the bot described in the FreqAI paper [36]. The FreqAI framework includes encapsulated AI techniques, enabling testing and comparing various time-series forecasting methods on typically chaotic and seemingly unpredictable cryptocurrency

price data. It is part of FreqTrade², which is highlighted as a versatile trading bot that allows users to define their strategies.

4 | Solution Design

Developing an arbitrage bot is essential to carry out algorithmic arbitrage trading. As discussed in Section 3, several arbitrage bots enable users to automate the entire arbitrage strategy. To stand out from competitors waiting for limited exploitable arbitrage opportunities, ML models are trained and integrated into the arbitrage bot to predict profitable arbitrages, considering confidence levels, before they materialize. The integration of ML principles is meant to push the algorithmization of the process of arbitrage trading forward and present a new approach to maximizing arbitrage profitability. The bot is implemented as a Python script that facilitates arbitrages between Binance and Bybit exchanges. These two centralized exchanges are chosen primarily due to their popularity among traders. Binance is currently the world's leading platform, known for the highest trading volume and core infrastructure services [20]. Bybit ranks as the second-largest exchange based on trading volume. Sharing similar objectives but with a smaller user base, Bybit is leveraged to exploit price margins between the two exchanges. The more traders are registered, the higher the amount of trades executed, possibly resulting in more significant price moves and more profitable arbitrages. For acquiring price data and implementing the required operations, the bot is linked to both exchanges through their APIs.³ The bot continuously searches for exploitable arbitrage opportunities in the BTCUSDT and ETHUSDT pairs until its operation is stopped. These two pairs are chosen due to their frequent trading, which can naturally result in higher arbitrages. The search is carried out on the futures market using limit orders, enhancing user protection in this predominantly low-risk strategy because the trade is not executed until the market reaches the target price. The maximal traded amount is limited to 0.01 BTC or 0.5 ETH with stop loss set to 10% to prevent high loss in case of incorrectly evaluated arbitrage possibility. Moreover, the risk management is covered by a timer for the open positions, limited to 1 min. Any position open for a more extended period without profit is closed to prevent losses. This corresponds to the nature of short-standing arbitrage trades that occur with high frequency; therefore, holding open positions for extended periods increases the risk of potential loss. For this, sufficient liquidity of all traded assets on Binance and Bybit is assumed. As a result, deposit or withdrawal fees are excluded to maximize potential gains and avoid diminishing the arbitrage opportunities with transaction costs.

Figure 1 presents the high-level execution flowchart, showing two approaches to seeking arbitrage opportunities. One involves a straightforward profit calculation, while the other incorporates ML. In both methods, the execution starts the same way. The API keys are requested, the connection with Binance and Bybit is established, and the cryptocurrency pairs for trading are collected together with the available balance. Afterward, the execution flow differs; if the ML model is included, the profitable arbitrage is predicted for the next time interval, and the trade is automatically executed after the pass of a one-time interval. Otherwise, the exact profit based on the current state of the order book is calculated, and the arbitrage trade is executed immediately. Anytime a trade is about to be executed, the available

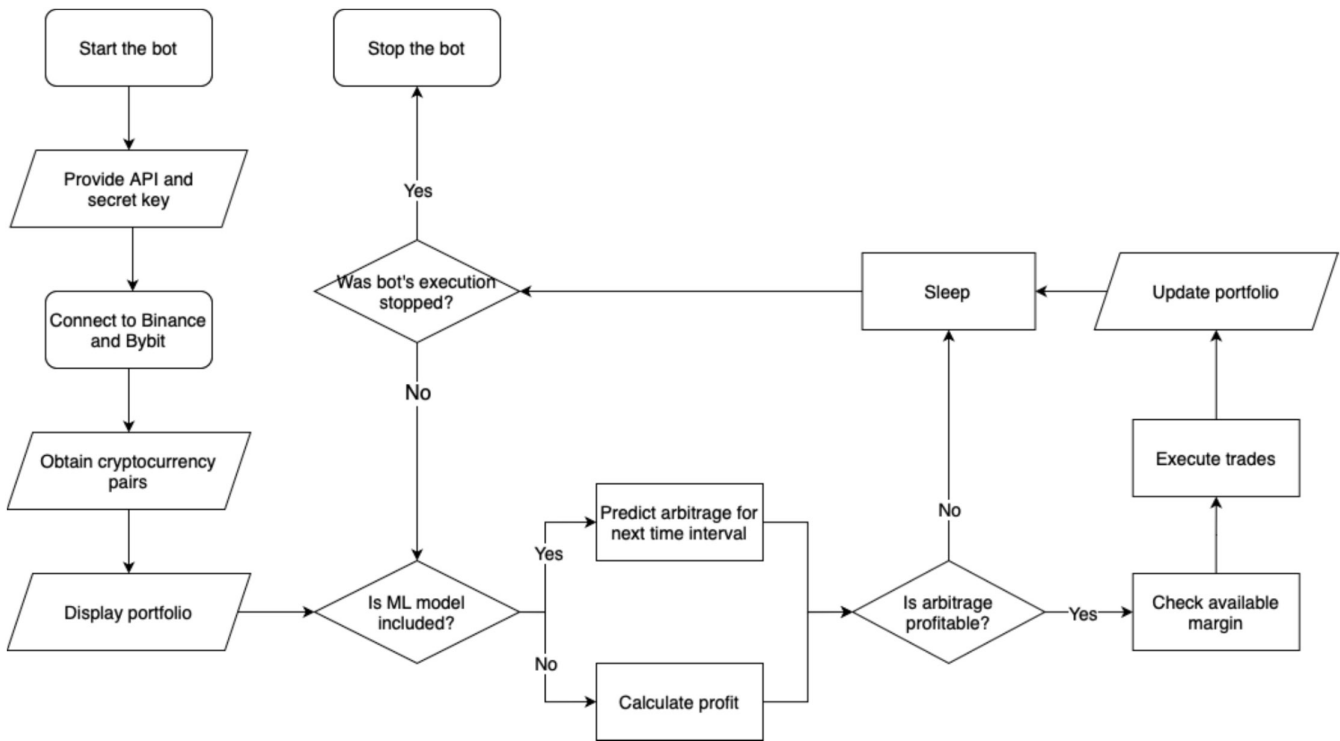


FIGURE 1 | Flowchart of arbitrage bot's execution.

margin is checked to protect the user from the potential loss of assets. After realizing the trades, the user's portfolio is updated, and the bot sleeps for 1 min before the search for a profitable arbitrage starts again.

To decide whether a purchase should be executed, the potential profit of the transaction sequence must be calculated. The current stage of the order book and transaction fees on both exchanges must be considered when considering profit in arbitrage trading. The calculation process consists of obtaining the traded amount of the base asset, obtaining the price difference defined by the bid-ask spread, and deducting transaction fees. To keep transaction fees as low as possible, all traded assets are held on both exchanges; thus, no transactions between exchanges are necessary, and naturally, no withdrawal or deposit fees are included. The calculation proceeds as follows [37]:

Traded amount of base asset

$$quote = quote / ((1 + fee) * price) \quad (1)$$

$$amount = \min(bid, ask, base, quote) \quad (2)$$

where:

bid = amount available on the bid side of the order book

ask = amount available on the ask side of the order book

base = amount of base assets available in the portfolio

quote = amount of quote assets available in a portfolio

fee = transaction fee of the exchange where buying

price = ask price from the order book

Transaction fees

$$fee = taker * amount * ask + taker * amount * bid \quad (3)$$

where:

taker = transaction fee as a percentage of the trade

amount = traded amount of a base asset

ask = ask price

bid = bid price

Potential profit

$$profit = |(ask - bid) * amount| - fee \quad (4)$$

$$percentage = profit / ask * 100 \quad (5)$$

where:

ask = ask price

bid = bid price

amount = traded amount of a base asset

fee = transaction fee of exchange where buying

percentage = percentage expression of profit

In addition to executing the described bot, all ML integration steps must be performed. Specifically, it consists of gathering data from both exchanges, data cleaning and preprocessing, visualization of the datasets, data normalization, and building the models for all algorithms with evaluation and comparison primarily based on the precision metric, supported by accuracy, area under the curve (AUC), and F1 score.

The datasets used to train the models consist of price data, specifically open, high, low, close, volume (OHLCV) data for the corresponding datetime. These values are proposed as a sufficient replacement for the actual state of the order book, and later, during the training, the open prices are utilized as bid and ask prices to simulate arbitrage predictability. Various time intervals, namely, 1, 5, and 15 min, are tested to find the interval that best reflects the cryptocurrency's dynamics. The shorter intervals are preferred due to the high volatility of the cryptocurrency market, as described in Section 2.3. The data are gathered for half a year to spot current trends in the market uninfluenced by the historical development of the currency. The datasets are gathered for Binance and Bybit separately to simulate potential arbitrages in the past.

As a part of the preprocessing, deletion of null values, handling of extreme outliers, and composition and appending of features representing change and arbitrage are executed. The change means the percentage movement of the currency since the last observation was collected. Arbitrage possibilities are expressed by 1 or 0, representing whether the observations from both exchanges for the specific datetime were likely to form an exploitable arbitrage.

5 | Evaluation

The arbitrage bot was implemented as proposed, continuously searching for arbitrage opportunities at 1-min intervals. To perform the required operations, the connection to the APIs of both exchanges is established through an API and secret keys corresponding to the created accounts. We have provided the new datasets on GitHub along with the previous paper source codes.⁴

The financial profit obtained strongly depends on the current stage of the cryptocurrency markets and world economics, seasonality, and time of day. The profitability of the bot was measured from 14.08 15:24:59 until 21.08 15:24:59, where the price of BTC on Binance fluctuated from 59160 to 59262.4 USDT and on Bybit from 59279.1 to 59220.70 USDT. The price of ETH on Binance fluctuated from 2657.27 to 2674.75 USDT and on Bybit from 2580.13 to 2635.36 USDT. The general change in the portfolio is shown in Table 2, resulting in a 191.072% increase in

the total balance of USDT on both exchanges. The success of the arbitrage bot is measured for all assets on both exchanges; the Binance column expresses the liquidity of the assets on the Binance exchange, and similarly, the liquidity is provided for the Bybit exchange. The assets are summed across all exchanges as no withdrawal or deposit transaction is executed to maximize the potential profit. This total liquidity of assets is then used to measure the arbitrage bot's total profit. The absolute change is the numerical difference between the amount of each asset before and after the execution of the arbitrage bot. The relative change represents the increase (or potentially decrease) of the total amount of USDT on both exchanges compared with the portfolio stage before and after the run of the arbitrage bot. When expressed in monetary values, the total amount of USDT increased from 25439.714 to 74047.849, and development proceeded as shown in Figure 2. This increase resulted in 77 transactions, where most crypto assets were bought on Bybit and sold on the Binance exchange. As a result of trading on the futures market, the total balance of BTC and ETH did not change. As already mentioned and also proven by the visualized execution, the arbitrage trading strategy results in profit when the purchases and sales are repeated over a more extended period without interruptions.

5.1 | ML Process

Including ML models required data gathering, data visualization, and preprocessing. The datasets obtained consisted of OHLCV data with a corresponding timestamp and volume traded for a half year, specifically from 05.02 until 06.08. Datasets were obtained for each pair traded on both exchanges for 1, 5, and 15 min. Due to the limitations of the APIs of both exchanges, the bid-ask spread must be estimated from the values of open and close prices. This is caused by limited or charged access to the historical states of the order book and executed trades. As part of preprocessing, the datasets were cleaned from null values and extreme outliers determined by values outside the quantile range 0.001 and 0.999. Later, the datasets were merged to concatenate the prices from both exchanges for every cryptocurrency pair and every time interval into one dataset and are described in more detail in Appendix D.

Furthermore, multiple columns were added, representing the percentage change between two open prices in a row and the identification of a probable arbitrage between exchanges. This probability is determined by statistical methods that evaluate whether the current stage of the market meets the conditions for an occurrence of a profitable arbitrage. For the evaluation, similar principles and calculations were used as described in Equations (3) and (4). The key difference lies in the considered

TABLE 2 | Portfolio after a run of arbitrage bot without ML for 1 week.

Binance	Bybit	Total	Absolute change	Relative change
0.017 BTC	0.65 BTC	0.667 BTC	0.000 BTC	0.000%
0.377 ETH	7.453 ETH	7.830 ETH	0.000 ETH	0.000%
15,734.444 USDT	58,313.405 USDT	74,047.849 USDT	48,608.135 USDT	191.072%

Visualization of total USDT profit through time

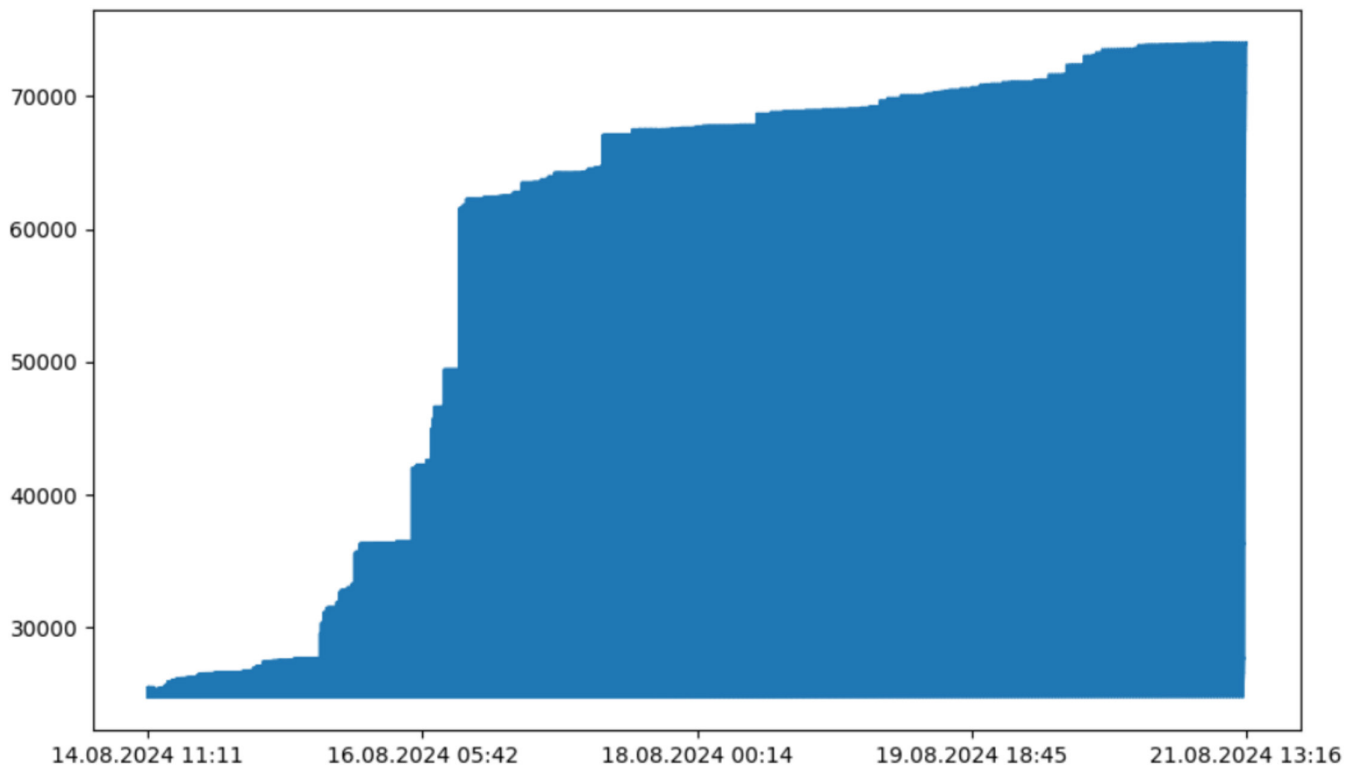


FIGURE 2 | The evolution of total amount of USDT during the execution of the arbitrage bot without ML. In percentage measure, the rise is equal to 191.072%, which corresponds to a value of 48608.135 USDT.

prices and traded amounts. Instead of bid and ask prices, the opening and closing prices from both exchanges are being used, providing a sufficient approximation to the direct calculation from the order book. The traded amount is set to constant values, as this amount does not determine the arbitrage conditions but only its height. These values are 0.01 for BTC and 1 for ETH, corresponding to the average value of executed trades. As part of the visualization, the changes on the exchanges were visualized aiming to demonstrate the potential differences suitable for arbitrages. Figure 3 shows the development of open prices and signifies that most price fluctuations resulting in arbitrage opportunities originate on the Bybit exchange, corresponding to the results of the execution of the arbitrage bot without ML. Similarly, Figure 4 visualizes the development of the prices on both exchanges using the traditional line structure. Figure 5 supports the price fluctuations visible from comparisons of open prices.

To enrich the arbitrage search, the datasets were divided into training and testing with a ratio of 80:20, and on the training set, multiple ML models were trained. Specifically, logistic regression, random forest, SVM, and MLP rely on the implementation provided by Python's scikit-learn library [38]. The training of each model consisted of iterations with individual combinations of hyperparameters and evaluating their impact on the testing set. Following this, the model possessing the optimal parameters is trained, and the most suitable time interval for the arbitrage prediction process is identified. Due to the higher number

of hyperparameters, large datasets for small time intervals, and negligible differences in individual tests, the set of all possible values for hyperparameters was not searched exhaustively. The hyperparameters that are the most appropriate for the paper's scenario are summarized in Table 3, with more details about the parameters' explanation, possible values, and corresponding values of metrics described in Appendix A. Based on the selected hyperparameters, the models from a technical perspective were structured as follows:

1. Logistic regression—For linear classification, a limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm (lbfgs) solver is used, further supported by Log loss with L2 regularization to prevent overfitting by applying the sum of the squares penalty to the model's coefficients. To use the regularization with an accurate strength, a medium value of 10 is assigned to the parameter C. Finally, to minimize the possibility of high variance and limit the execution time, the maximum number of allowed iterations is set to 500.
2. Random forest—The model makes its predictions by combining outputs from multiple decision trees, also called estimators, whose number is limited to 150. As a criterion to determine a split inside a single node, the Gini measure is the impurity of samples in the specific leaf. This splitting criterion is only applied to leaves with more than 10 samples as a method for overfitting prevention. On top of

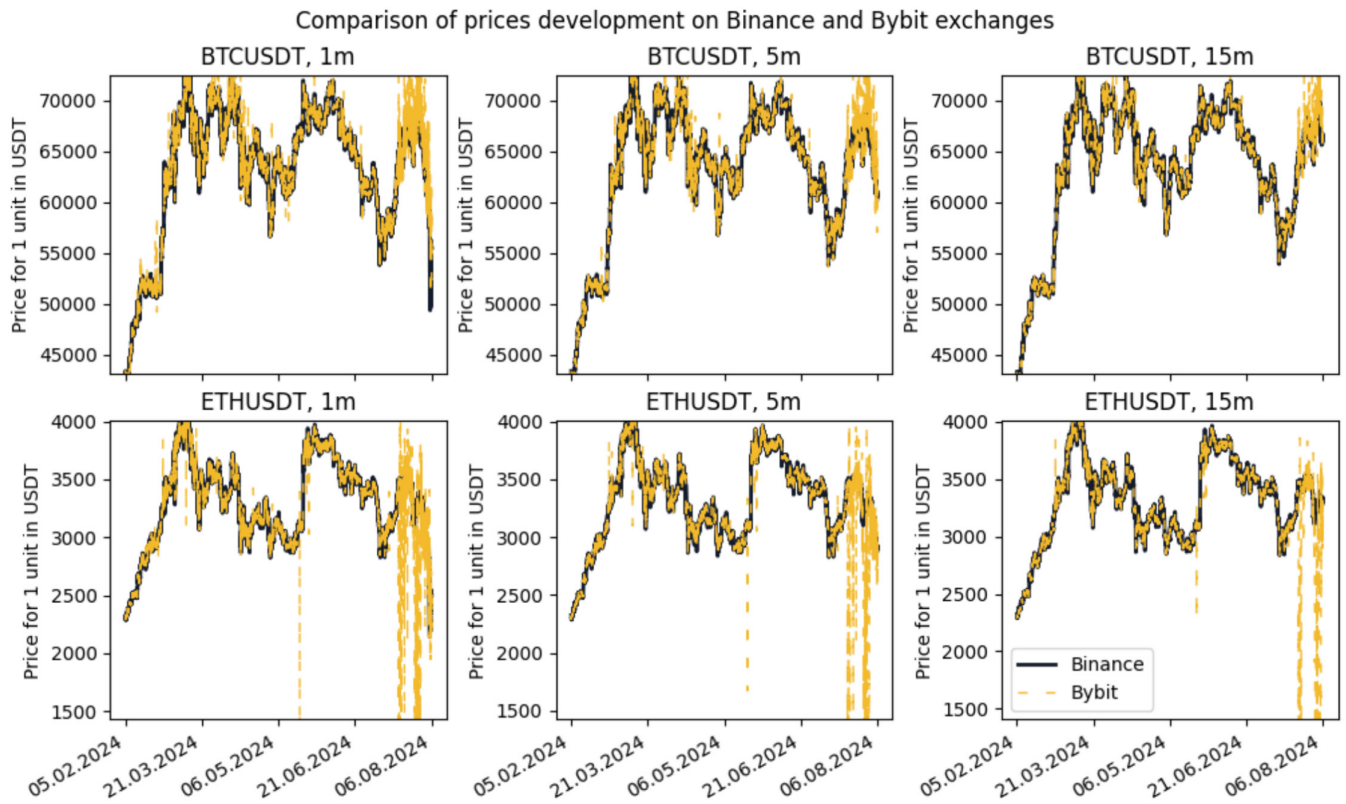


FIGURE 3 | The evolution of cryptocurrency prices. It compares exactly two cryptocurrency pairs, BTCUSDT and ETHUSDT, with price values obtained between 05.02 and 06.08. The prices are collected at different intervals, namely, 1-, 5-, and 15-min intervals, from the two exchanges, Binance and Bybit. The individual graphs compare the open prices from both exchanges for one cryptocurrency pair and one-time intervals to visualize the price difference and show potentially profitable arbitrage.

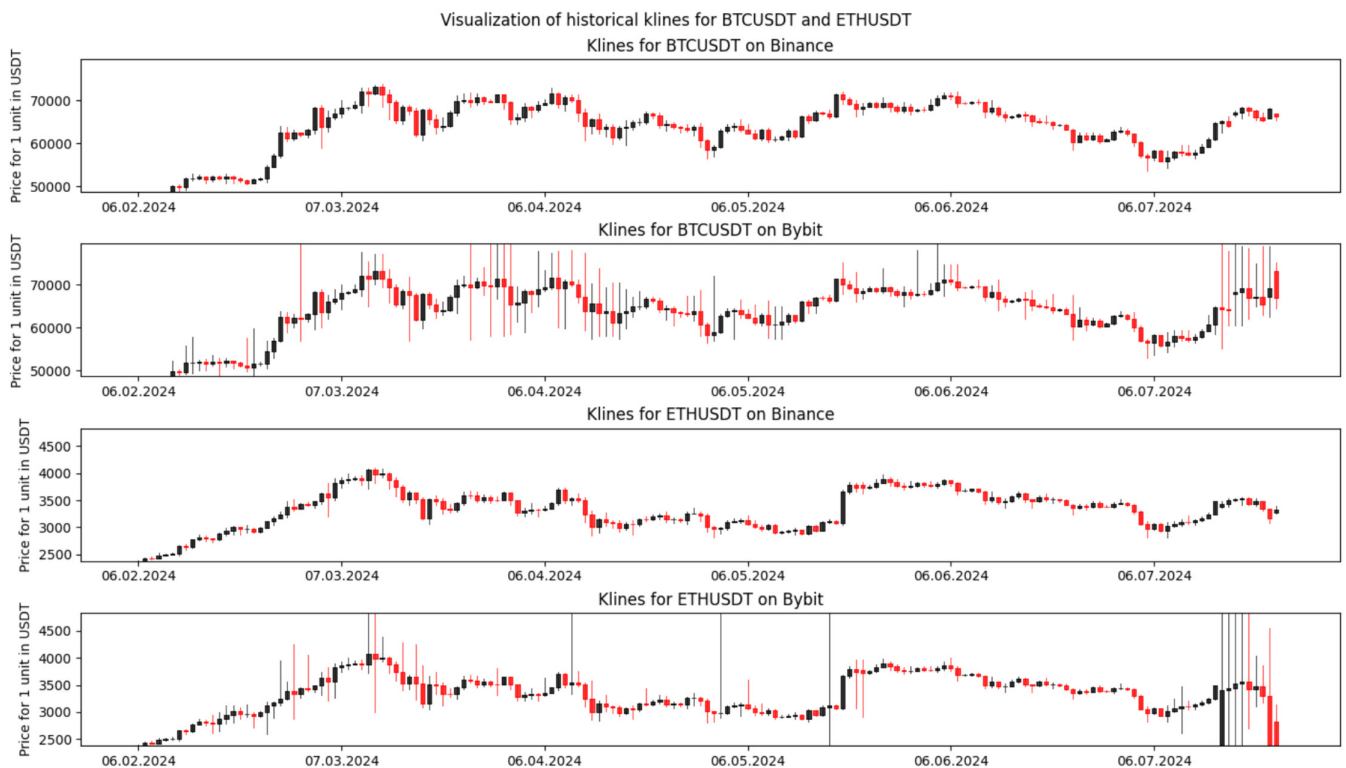


FIGURE 4 | The evolution of cryptocurrency prices. Here, klines visualization demonstrates the price movements for both cryptocurrency pairs, namely, BTCUSDT and ETHUSDT, on Binance and Bybit. Compared with the previous graph, here, the OHLCV data are utilized only for 5-min intervals. The data were gathered from 05.02 to 06.08 and visualize the historical development of the considered cryptocurrency pairs.

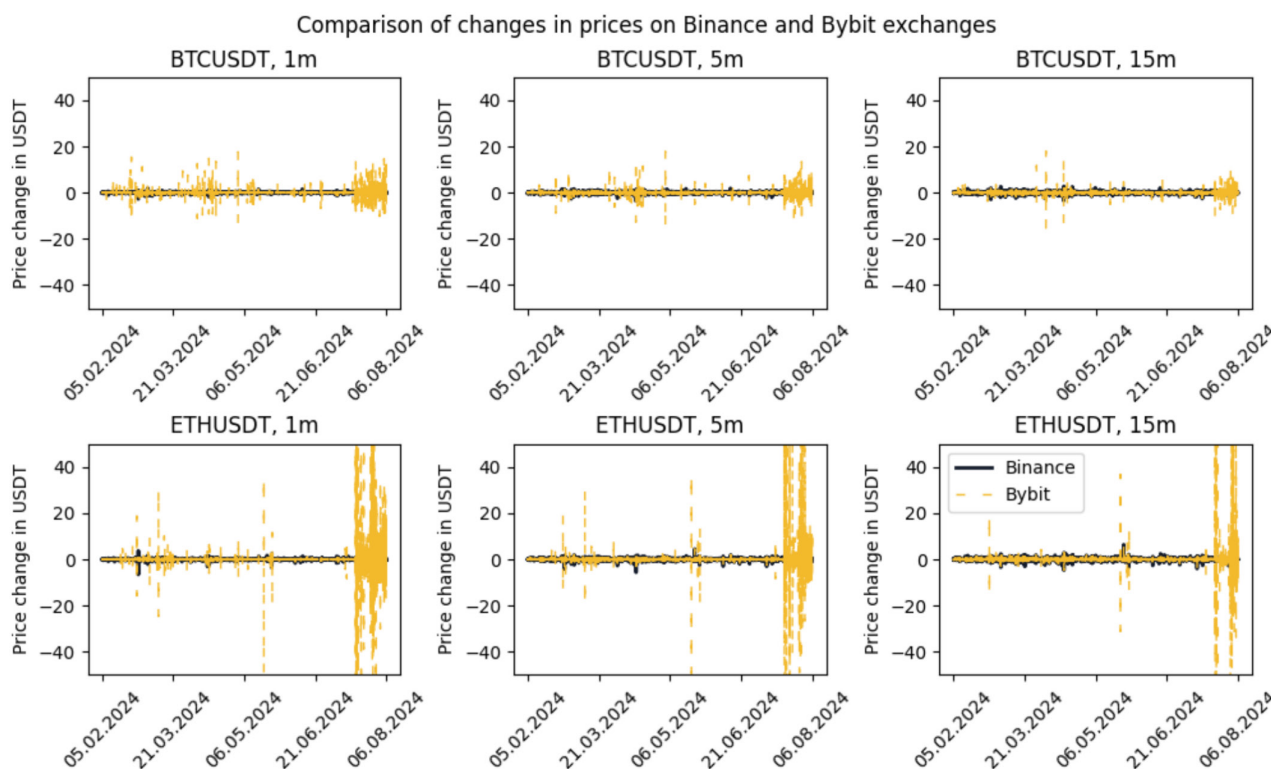


FIGURE 5 | The arbitrage opportunities and the calculation of the change between two consecutive absolute values of open prices. The higher this change becomes, the more likely there is an arbitrage opportunity to take advantage of unaligned price changes in the cryptocurrency markets.

TABLE 3 | Hyperparameters for machine learning models.

ML model	Best hyperparameters				
Logistic regression	'penalty': 'l2'	'C': 10	'max_iter': 500		
Random forest	'n_estimators': 150	'criterion': 'gini'	'min_samples_split': 10	'max_samples_split': 'sqrt'	'bootstrap': True
Support vector machine	'C': 10	'max_iter': 1500	'kernel': 'linear'		
Multilayer perceptron	'activation': 'relu'	'solver': 'sgd'	'learning_rate_init': 0.01	'max_iter': 200	'hidden_layer_sizes': (5, 5)

this, the number of used features is equal to the square root of their total amount, and only the most valuable features are selected. This feature selection is left to decide for the model provided by scikit-learn. The highest value of the contribution of the feature is taken into account, and as the most valuable based on this criterion, the features analyzing the Bybit exchange were selected. As the most frequently chosen, the values for open, low, and close price and traded volume all occurred on Bybit.

3. SVM—The model provides the best results with a linear kernel, so adding more dimensions to the data does not help the model fit more precisely. Similar to logistic regression, a medium value of the C parameter is used to accurately adjust the regularization strength. Further, more iterations are necessary for this model to be trained, and limited to 1500.

4. MLP—The structure of this model has two hidden layers, each with five neurons and an output with two neurons for every predictable class. Based on the default definition of the scikit-learn library, the model uses rectified linear unit (ReLU) as an activation function for the hidden layers. Further, a logistic sigmoid function is specified as an activation function for the output layer, and in order to find the most appropriate value of the loss function a stochastic gradient descent (SGD) as an optimization method is used. The learning rate is set to the medium value of 0.01, and the maximum number of iterations is limited to 200.

Further, as arbitrages can be unpredictable, the model gives us confidence in its prediction. This confidence is represented as the value of the model's outcome before the last activation function, which is responsible for classifying the output into one of the

available classes (occurrence of arbitrage or its absence). These values can be interpreted as confidence in the model for positive and negative predictions. This confidence is converted into binary class outcomes based on a specified threshold. The default value of this threshold is determined by the Youden index calculated from the receiver-operating characteristic (ROC) curve. Further, the default threshold calculated by the Youden index is taken as guidance. The experiments were conducted on 11 values with a small distance from this threshold. Specifically, five smaller and five larger values than the default threshold were tested, each with a difference of 0.03 from the previous value. This difference in value is appropriate because we want to stay close to the default threshold but try out other possibilities, too.

The comparison of the suitability of individual values is based on the potential profit obtained by executing all predicted arbitrages throughout the whole test dataset. The corresponding thresholds with the highest profitability are summarized in Figure 6 separately for every model, time interval, and asset pair. Generally, the best models from the metrics perspective have one of the highest thresholds, meaning that higher confidence in the occurrence of profitable arbitrage leads to higher profit. The models are more prone to being careful and require more confidence to predict an arbitrage opportunity.

The results are gathered primarily based on the precision and supported by the proportion of accuracy, AUC, and F1 score with

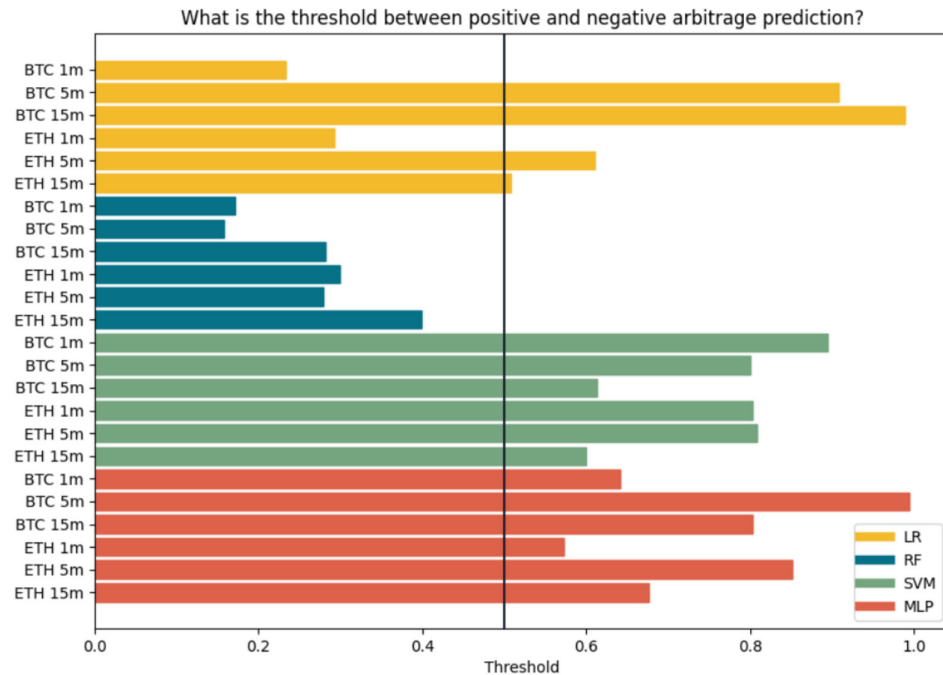


FIGURE 6 | The comparison of the best thresholds for individual models for every cryptocurrency pair and every time interval. The default threshold of 0.5 is marked, and every bar below this value is more benevolent in the prediction of arbitrage. On the opposite side of the default value, the models are more careful and predict arbitrage only with a higher confidence level. Due to the limited space, the models and traded pairs are markets with shortcuts with the following meaning: LR = logistic regression, RF = random forest, SVM = support vector machine, MLP = multilayer perceptron, BTC = BTCUSDT, ETH = ETHUSDT.

TABLE 4 | ML models.

ML model	Cryptocurrency	Time interval	Precision	AUC score	F1 score	Accuracy
Logistic regression	BTCUSDT	15 min	0.091	0.621	0.133	0.988
	ETHUSDT	5 min	0.794	0.972	0.868	0.986
Random forest	BTCUSDT	1 min	0.211	0.914	0.337	0.994
	ETHUSDT	5 min	0.802	0.974	0.874	0.987
Support vector machine	BTCUSDT	1 min	0.075	0.56	0.093	0.996
	ETHUSDT	15 min	0.561	0.932	0.69	0.966
Multilayer perceptron	BTCUSDT	5 min	0.333	0.638	0.303	0.998
	ETHUSDT	5 min	0.82	0.968	0.879	0.987

Note: The summarization of the trained ML models. Precision metrics, AUC score, F1 score, and accuracy are provided for every model. The summary highlights only the best time interval per ML model and cryptocurrency pair, and the overall best model for every cryptocurrency pair is marked in red. All the evaluation metrics for every ML model, cryptocurrency pair, and time interval are described in depth in Appendix B.

the most suitable time interval for each pair of cryptocurrencies, summarized in Table 4. These results are supported by the ROC curves shown in Figures 7 and 8 that visualize the model's performance across all thresholds during the training. As the cryptocurrency markets are characterized by high volatility, the models must be retrained to keep up with the market trends. Currently, the models are being retrained once per month, but it could be beneficial for the model to be retrained near real-time. This would allow us to learn from the executed trades and the current development of the prices, and consequently follow the volatile trend of the markets.

5.2 | Arbitrage Bot With ML

Subsequently, due to better results, the ETHUSD prediction model was included in the execution of the arbitrage bot. The profitability was measured again on 23.08 10:46:34 to 30.08 10:46:34, where the price of ETH on Binance fluctuated from 2686 to 2518.97 USDT and on Bybit from 2691.81 to 2504.75 USDT. A different date range is chosen for the implementation without ML, as both these versions are being run on the same cryptocurrency exchanges and, therefore, would steal opportunities from each other, and it would not be possible to prove the efficiency of any of them. The general change in the portfolio is shown in Table 5, resulting in a 199.233% change in the total balance of USDT on both exchanges, with the same meaning of the column values as in the previous portfolio table. When expressed in monetary values, the total amount of USDT increased from 62125.841 to 185901.019, and development proceeded as shown in Figure 9. This increase results from 389 transactions, in which the crypto asset was bought on Bybit and sold on the

Binance exchange in all cases. As can be seen from the results, the benefit of incorporating ML is significant. However, the possibility of volatility affecting the exact increase in the total amount of assets must be considered, as the bot will likely make a higher profit in periods of higher volatility. The model can be more profitable when predicting the occurrence of arbitrage for the upcoming 5-min interval using the MLP architecture. Also, fewer transactions achieve higher profits, meaning the model finds more profitable arbitrage opportunities. The advantage of ML inclusion could be particularly advantageous as the cryptocurrency market matures and becomes less volatile. This causes the number of arbitrage opportunities to decrease, and the ability to predict it can help to outrun rivals. There are numerous causes of the misalignment between predictions and the actual state of the market. Firstly, the cryptocurrency market is still considered volatile and inefficient, and any predictions are complicated. Moreover, sources of arbitrage possibilities are plentiful, and not all of them can be captured by a single ML model, which is also not errorless.

5.3 | Hypothesis Testing

Two hypotheses were stated concerning a cryptocurrency pair's time interval and volatility to prove the statistical significance of the obtained results and their impact on the analyzed data.

1. Hypothesis for time interval

- Null hypothesis: 1-min time intervals have the same precision as other intervals when predicting an exploitable arbitrage.

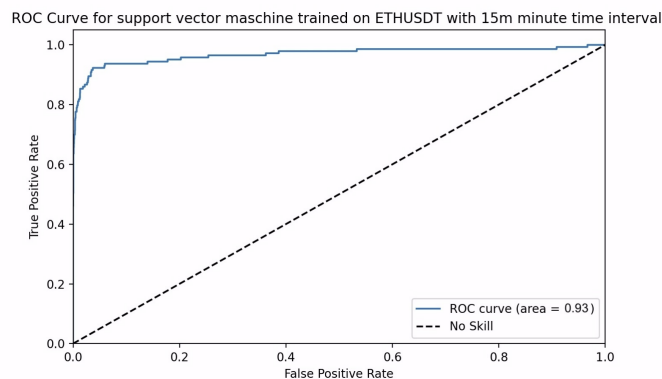
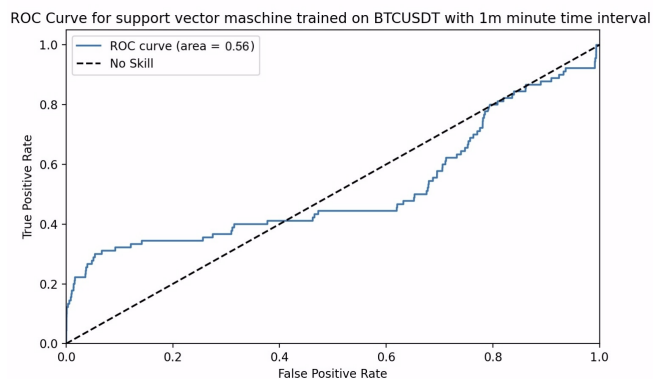
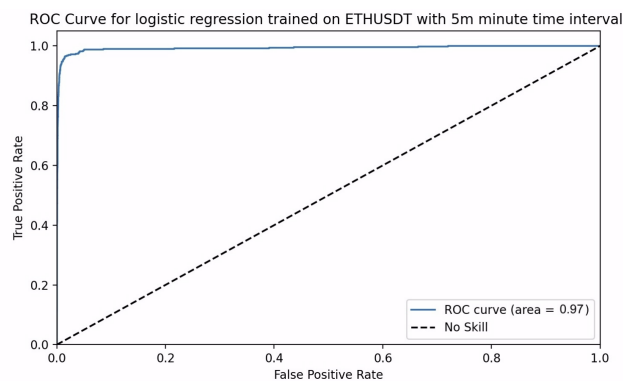
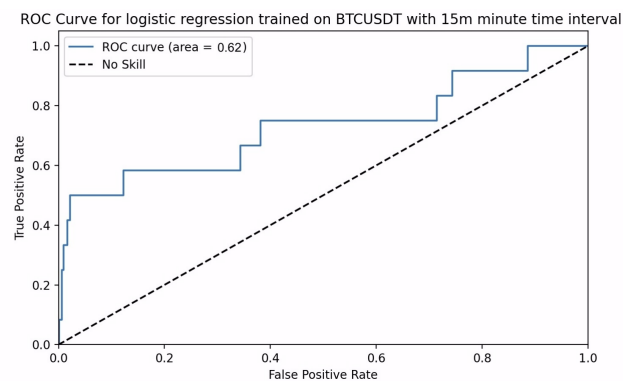


FIGURE 7 | The ROC curves for the best results by the precision metrics for every machine learning model and cryptocurrency pair—Part 1.

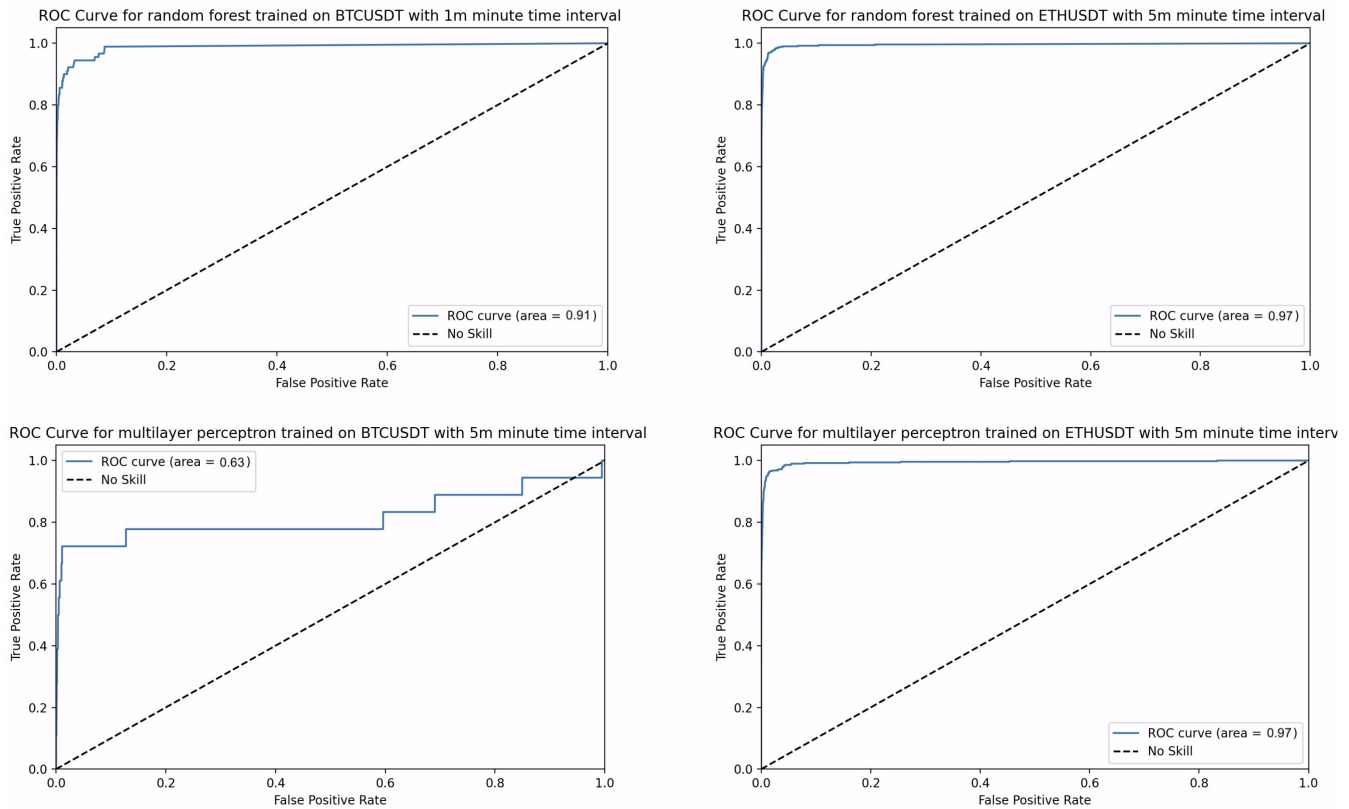


FIGURE 8 | The ROC curves for the best results by the precision metrics for every machine learning model and cryptocurrency pair—Part 2.

TABLE 5 | Portfolio after run of arbitrage bot with ML for 1 week.

Binance	Bybit	Total	Absolute change	Relative change
0.017 BTC	0.650 BTC	0.667 BTC	0.000 BTC	0.000%
0.377 ETH	7.453 ETH	7.830 ETH	0.000 BTC	0.000%
15,297.046 USDT	170,603.973 USDT	185,901.019 USDT	123,775.178 USDT	199.233%

- Alternative hypothesis: 1-min time intervals yield higher precision than other intervals when predicting exploitable arbitrage.

2. Hypothesis for the volatility of a cryptocurrency pair

- Null hypothesis: Arbitrage opportunities between less volatile cryptocurrencies could be predicted with the same precision.
- Alternative hypothesis: Arbitrage opportunities between less volatile cryptocurrencies could be predicted more precisely.

The samples used for the testing consisted of precision values obtained from testing all ML models on all datasets available. Specifically, for the testing of time interval impact, precision values were divided based on the time intervals of the datasets, namely, 1-, 5-, and 15-min intervals. For the volatility hypothesis, the obtained values were classified by the cryptocurrency pair that the dataset represents. Specifically, BTCUSDT is more

volatile, and ETHUSDT is less volatile. Afterward, the collected samples were tested for the independence of observations, homogeneity of the variance, and normality of the data.

The proper statistical tests were chosen based on the stated hypotheses' characteristics. The one-way ANOVA test (or its nonparametric variation, Kruskal–Wallis H test) was applied for the first hypothesis with three different categories of predictors. The second hypothesis was tested by an independent *t*-test (or its nonparametric variation, the Wilcoxon rank-sum test) because the outcome variables come from different populations. All tests were implemented with a 0.05 significance level.

Afterward, both hypotheses were tested. As assumptions about the normal distribution of the data were not met, both hypotheses were tested by the proposed nonparametric versions of the tests. All statistical tests performed gave the results summarized in Table 6 where *p*-values are provided. The *p*-values that result in rejecting the null hypothesis are highlighted to emphasize the results necessary for this research. It is shown that by

Visualization of total USDT profit through time

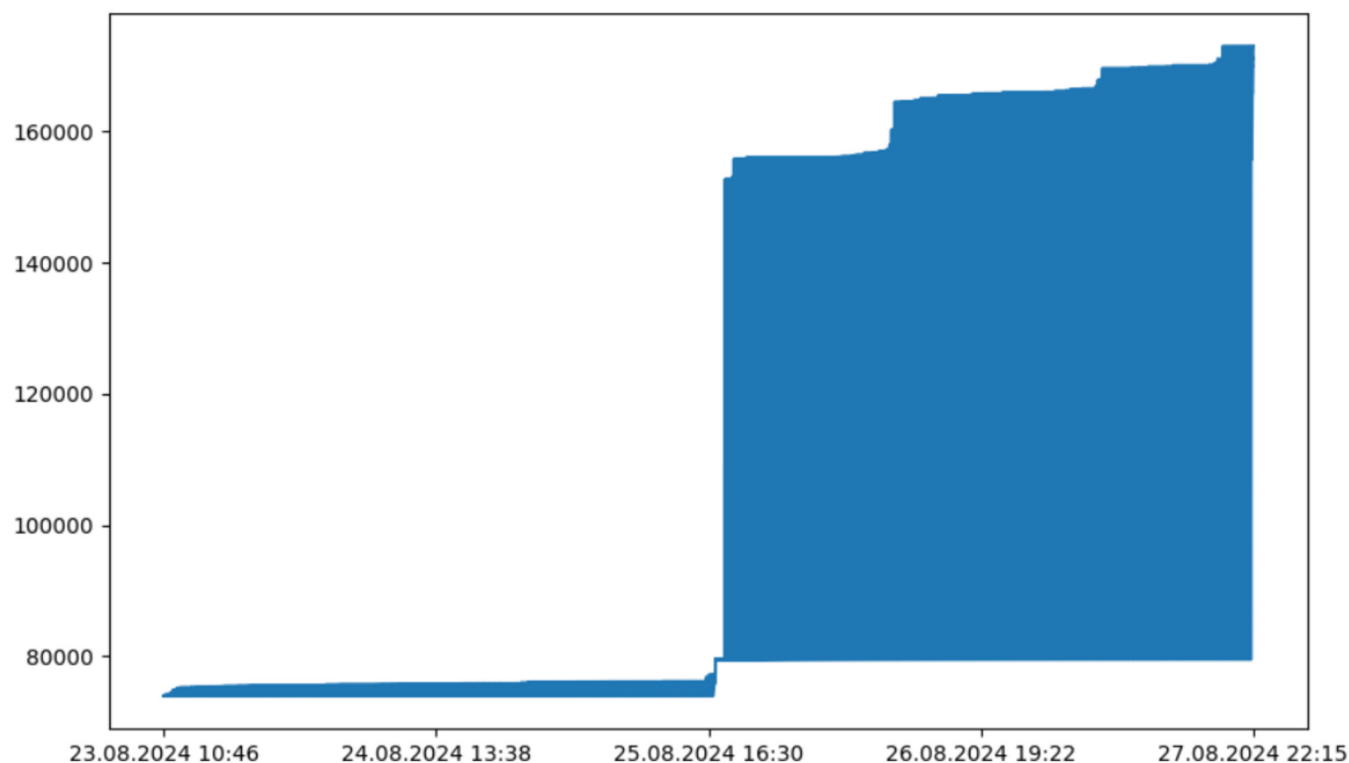


FIGURE 9 | The evolution of total amount of USDT during the execution of the arbitrage bot with ML. In percentage measure, the rise is equal to 199.233%, which corresponds to a value of 123775.178 USDT.

TABLE 6 | Statistical tests.

	Independence test	Variance test	Normality test	Nonparametric test
Time interval	0.9999	0.8219	0.0733	0.8698
			0.0561	
			0.0461	
Volatility	0.9999	0.1618	0.0469	4.146e-05
			0.0046	

Note: For the testing of the variables' independence, the chi-squared test was used; for the equal variance verification, the Levene test was applied; normality was tested via the Shapiro–Wilk test, and as a nonparametric test for the time interval hypothesis, the Kruskal–Wallis H test, and for the volatility hypothesis, the Wilcoxon rank-sum test were utilized.

the first hypothesis, we failed to reject the null hypothesis, as no statistical difference between the individual time intervals was detected. This means that the 1-min time interval (or any other utilized time interval) did not significantly impact the precision. Therefore the suitability of a specific time interval has to be determined purely by testing of ML models. The second hypothesis, with a p -value below the significance level, causes a rejection of the null hypothesis. This can be interpreted as statistically significant evidence that less volatile cryptocurrencies can be predicted with higher precision. To sum up, volatility significantly impacts the evaluation of ML models by precision metrics; however, the relationship between a 1-min time interval and precision is limited.

5.4 | Predictions on Stable and Volatile Markets

As the amount of arbitrage profit is highly dependent on the price moves, the market's volatility has to be considered as an impact factor. On the other hand, non-negative profitability can be obtained regardless of this factor. Compared with the previous version [12] tested during a more stable period, the results provided during a more volatile period naturally offer higher profitability. When expressed numerically, the percentage change moved from 32.11 % (total amount of USDT rose from 26,364.72 to 34,830.39) to 191.072 % (total amount of USDT increased from 25,439.714 to 74,047124.849) for the arbitrage bot without ML and from 0.455 % (total amount of USDT rose from 34,730.91 to

34,888.94) to 199.233% (total amount of USDT increased from 62125.841 to 185,901.019) for the arbitrage bot with ML integration. The fact causes this: The price deviations between the exchanges are higher and occur more often. However, the higher profit obtained is supported by the adequate addressing of risk management, which has a significant impact, especially on the profitability of the arbitrage bot with ML integration. As described above, this risk management includes limits on the maximum amount of traded assets and the time interval for holding a position opened together with stop loss.

6 | Discussion

The results of the training of ML models signify a relationship between OHLCV price data of a cryptocurrency pair, traded volume, percentage change, and the occurrence of exploitable arbitrage. When considering the differences between individual ML models, the best results were obtained from MLP, but being aware of the potential overfitting and thus preferring a 5-min time interval, as a 1-min interval is prone to learning the dataset. The main difference was identified between the considered cryptocurrency pairs. The more volatile ones are much harder to predict, regardless of the time interval. As Figure 3 shows, the pair ETHUSDT is much more stable for the selected time range of the dataset, resulting in significantly better results for all considered metrics for the implemented models. On the other hand, less volatile cryptocurrencies create fewer arbitrage opportunities, resulting in lower available profit. When all combined, MLP for a 5-min time interval and trained on the ETHUSDT cryptocurrency pair leads to the best evaluation with the precision of 0.82, AUC scores 0.968, F1 score 0.879, and accuracy of 0.987.

The paper focused on integrating ML algorithms into an arbitrage trading strategy between cryptocurrency exchanges. From all the models described above, MLP performed the best when used for a 5-min time interval in the dataset. When integrated into a real trading strategy, the approaches with and without the inclusion of the ML model were compared, resulting in a profit of 191.072% without and 199.233% with the ML model. Although these results were obtained in different date ranges, both numbers support the profitable modifications to the implemented arbitrage bot.

Compared with related work, the obtained results were similar in some aspects but different in others. Equally, to T. G. Fischer et al. [6], random forest outperformed logistic regression; however, with a dataset consisting of nearly the same price data, it thrived better with a 5-min interval in comparison to the 1-min interval from the related work. Similarly, Madan et al. [33] considered random forest more successful than logistic regression, whose results were not influential in this paper. This difference could be explained by the diametrically different features and time intervals considered. However, incorporating simple deep learning techniques into arbitrage predictability is relatively rare, and as shown by our results, it could be a valid possibility.

7 | Conclusion and Future Work

To sum up, it was demonstrated that combining ML algorithms and an arbitrage trading strategy between cryptocurrency

exchanges can provide interesting outcomes. Multiple ML models with various parameters were tested. MLP with a 5-min time interval provided the best result for both considered cryptocurrency pairs: BTCUSDT and ETHUSDT. Both approaches to arbitrage trading with and without the inclusion of ML algorithms were tested and provided profitable results. The key novelty of the paper is the profitability demonstration of the prediction of exploitable arbitrage one-time intervals beforehand, with confidence level integration to predict an arbitrage only in case of higher certainty of the model's predictions. All these innovations provide a valuable resource to outrun the rivals represented by an increasing number of existing arbitrage bots and decrease the volatility of maturing cryptocurrency markets.

Furthermore, to obtain the maximal relationships from the datasets, the possibility of arbitrage occurrence appended to each dataset could be replaced by observing whether trades leading to arbitrage were executed by obtaining a historical order book and bid-ask spread from the involved exchanges. Such an analysis of data could lead to more precise results, even in volatile crypto markets. Including less-known, more unstable currencies could also influence the results due to frequent arbitrage possibilities, making it harder to predict exploitable arbitrage. Implementing more complex NNs could be beneficial for revealing the relationships between the features in datasets. We plan to focus on arbitrage predictions on decentralized exchanges (DEXs), promising more opportunities. Further, connecting data like crypto markets changes [39] based on the information from market makers, could even more higher the profitability.

Acknowledgments

Funded by the EU NextGenerationEU through the Recovery and Resilience Plan for Slovakia under the Project No. 09I05-03-V02-00057. This work was supported by the Science Grant Agency—Project VEGA 1/0300/25.

Funding

This work was supported by EU NextGenerationEU (09I05-03-V02-00057) and Vedecká Grantová Agentúra MŠVVaŠ SR a SAV (1/0300/25).

Data Availability Statement

The data that support the findings of this study are openly available in ML-for-arbitrage-in-cryptoexchanges at <https://github.com/fiit-ba/ML-for-arbitrage-in-cryptoexchanges/>.

Endnotes

¹“Market efficiency refers to how well current prices reflect all available, relevant information about the actual value of the underlying assets. A truly efficient market eliminates the possibility of beating the market because any information available to any trader is already incorporated into the market price.”—Investopedia.

²{<https://github.com/freqtrade/freqtrade>}

³Query format is influenced by the following repositories: <https://github.com/CryptoFacilities/REST-v3-Python/blob/master/cfRestApiV3.py>, https://github.com/bybit-exchange/api-usage-examples/blob/master/V3_demo/api_demo/contract/Encryption_HMAC.py.

⁴<https://github.com/fiit-ba/ML-for-arbitrage-in-cryptoexchanges>.

References

1. R. Levus, A. Berko, L. Chyrun, and V. Panasyuk, "Intelligent System for Arbitrage Situations Searching in the Cryptocurrency Market," In *MoMLeT+ DS*, 2021, p. 407–440.
2. G. Peduzzi, J. James, and J. Xu, "Jack the Rippler: Arbitrage on the Decentralized Exchange of the XRP Ledger," in 2021 3rd Conference on Blockchain Research & Applications for Innovative Networks and Services (BRAINS), Sep. 2021, 1–2, <https://doi.org/10.1109/BRAINS52497.2021.9569833>.
3. V. Lytvyn, V. Vysotska, V. Kuchkovskiy, et al., "Development of the System to Integrate and Generate Content Considering the Cryptocurrency Needs of Users," *Eastern-European Journal of Enterprise Technologies* 1, no. 2 (2019): 18–39, <https://doi.org/10.15587/1729-4061.2019.154709>.
4. L. Zhou, K. Qin, and A. Gervais, "A2MM: Mitigating Frontrunning, Transaction Reordering and Consensus Instability in Decentralized Exchanges," arXiv, Jun. 17, 2021, <http://arxiv.org/abs/2106.07371>.
5. J. Chen, D. Lin, and J. Wu, "Do Cryptocurrency Exchanges Fake Trading Volumes? An Empirical Analysis of Wash Trading Based on Data Mining," *Physica A: Statistical Mechanics and Its Applications* 586 (2022): 126405, <https://doi.org/10.1016/j.physa.2021.126405>.
6. T. G. Fischer, C. Krauss, and A. Deinert, "Statistical Arbitrage in Cryptocurrency Markets," *Journal of Risk and Financial Management* 12, no. 1 (2019): 1, <https://doi.org/10.3390/jrfm12010031>.
7. P. Schober and T. R. Vetter, "Logistic Regression in Medical Research," *Anesthesia and Analgesia* 132, no. 2 (2021): 365–366, <https://doi.org/10.1213/ANE.0000000000005247>.
8. E. Akyildirim, A. Goncu, and A. Sensoy, "Prediction of Cryptocurrency Returns Using Machine Learning," *Annals of Operations Research* 297, no. 1 (2021): 3–36, <https://doi.org/10.1007/s10479-020-03575-y>.
9. M. Schonlau and R. Y. Zou, "The Random Forest Algorithm for Statistical Learning," *Stata Journal* 20, no. 1 (2020): 3–29, <https://doi.org/10.1177/1536867X20909688>.
10. D. A. Pisner and D. M. Schnyer, "Chapter 6—Support Vector Machine," in *Machine Learning*, eds. A. Mechelli and S. Vieira (Academic Press, 2020), 101–121, <https://doi.org/10.1016/B978-0-12-815739-8.00006-7>.
11. H. Taud and J. F. Mas, "Multilayer Perceptron (MLP)," in *Geomatic Approaches for Modeling Land Change Scenarios. Lecture Notes in Geoinformation and Cartography*, eds. M. T. C. Olmedo, M. Paegelow, J.-F. Mas, and F. Escobar (Springer International Publishing, 2018), 451–455, https://doi.org/10.1007/978-3-319-60801-3_27.
12. K. Okasová and K. Košťál, "Using Machine Learning for Predicting Arbitrage Occurrences in Cryptocurrency Exchanges," 2024 IEEE International Conference on Blockchain and Cryptocurrency (ICBC), Dublin, Ireland, 2024, 1–7, <https://doi.org/10.1109/ICBC59979.2024.10634339>.
13. E. Barucci, G. Giuffrè Moncayo, and D. Marazzina, "Market Impact and Efficiency in Cryptoassets Markets," *Digital Finance* 5, no. 3 (2023): 519–562, <https://doi.org/10.1007/s42521-023-00095-9>.
14. S. Krückeberg and P. Scholz, "Decentralized Efficiency? Arbitrage in Bitcoin Markets," *Financial Analysis Journal* 76, no. 3 (2020): 135–152.
15. E. Han, "Intra-Exchange Cryptocurrency Arbitrage Bot," Master of Science, San Jose State University, San Jose, CA, USA, (2018), <https://doi.org/10.31979/etd.6xze-y9xu>.
16. S. Bistarelli, A. Cretarola, G. Figà-Talamanca, and M. Patacca, "Model-Based Arbitrage in Multi-Exchange Models for Bitcoin Price Dynamics," *Digital Finance* 1, no. 1–4 (2019): 23–46, <https://doi.org/10.1007/s42521-019-00001-2>.
17. T. Hattori and R. Ishida, "The Relationship Between Arbitrage in Futures and Spot Markets and Bitcoin Price Movements: Evidence From the Bitcoin Markets," *Journal of Futures Markets* 41, no. 1 (2021): 105–114, <https://doi.org/10.1002/fut.22171>.
18. A. Kabašinskas and K. Štutienė, "Key Roles of Crypto-Exchanges in Generating Arbitrage Opportunities," *Entropy* 23, no. 4 (2021): 455, <https://doi.org/10.3390/e23040455>.
19. H. Gupta and R. Chaudhary, "An Empirical Study of Volatility in Cryptocurrency Market," *Journal of Risk and Financial Management* 15, no. 11 (2022): 11, <https://doi.org/10.3390/jrfm15110513>.
20. M. S. Ahmed, A. A. El-Masry, A. I. Al-Maghyereh, and S. Kumar, "Cryptocurrency Volatility: A Review, Synthesis, and Research Agenda," *Research in International Business and Finance* 71 (2024): 102472, <https://doi.org/10.1016/j.ribaf.2024.102472>.
21. M. Bhatnagar, S. Taneja, and R. Rupeika-Apoga, "Demystifying the Effect of the News (Shocks) on Crypto Market Volatility," *Journal of Risk and Financial Management* 16, no. 2 (2023): 2, <https://doi.org/10.3390/jrfm16020136>.
22. V. L. Tran and T. Leirvik, "Efficiency in the Markets of Cryptocurrencies," *Finance Research Letters* 35 (2020): 101382, <https://doi.org/10.1016/j.frl.2019.101382>.
23. F. Aslam, B. A. Memon, A. I. Hunjra, and E. Bourri, "The Dynamics of Market Efficiency of Major Cryptocurrencies," *Global Finance Journal* 58 (2023): 100899, <https://doi.org/10.1016/j.gfj.2023.100899>.
24. A. Morin and T. Moore, "The Impact of Cryptocurrency Spot Price Volatility on Derivative Markets," Feb. 17, 2024, Rochester, NY: 4729600, <https://doi.org/10.2139/ssrn.4729600>.
25. J. Liu and V. J. Valcarcel, "Hedging Inflation Expectations in the Cryptocurrency Futures Market," *Journal of Financial Stability* 70 (2024): 101205, <https://doi.org/10.1016/j.jfs.2023.101205>.
26. R. D. Blasis and A. Webb, "Arbitrage, Contract Design, and Market Structure in Bitcoin Futures Markets," *Journal of Futures Markets* 42, no. 3 (2022): 492–524, <https://doi.org/10.1002/fut.22305>.
27. B. Łet, K. Sobański, W. Świder, and K. Włosik, "What Drives the Popularity of Stablecoins? Measuring the Frequency Dynamics of Connectedness Between Volatile and Stable Cryptocurrencies," *Technological Forecasting and Social Change* 189 (2023): 122318, <https://doi.org/10.1016/j.techfore.2023.122318>.
28. K. Mokni, G. El Montasser, A. N. Ajmi, and E. Bourri, "On the Efficiency and Its Drivers in the Cryptocurrency Market: The Case of Bitcoin and Ethereum," *Financial Innovation* 10, no. 1 (2024): 39, <https://doi.org/10.1186/s40854-023-00566-3>.
29. P. Katsiampa, "An Empirical Investigation of Volatility Dynamics in the Cryptocurrency Market," Jun. 25, 2018, Rochester, NY: 3202317, <https://doi.org/10.2139/ssrn.3202317>.
30. D. G. Baur and T. Dimpfl, "Asymmetric Volatility in Cryptocurrencies," Sep. 18, 2018, Rochester, NY: 3347617, <https://doi.org/10.2139/ssrn.3347617>.
31. H. P. Winotoatmojo, A. A. Setyawan, A. R. Hendraningrat, and J. G. Setiawati, "Cryptocurrency Market Dynamics: Analyzing Trends and Patterns in Bitcoin," *Dinasti International Journal of Education Management and Social Science* 5, no. 4 (2024): 558–564, <https://doi.org/10.31933/dijemss.v5i4.2553>.
32. R. Auer, G. Cornelli, S. Doerr, et al., "Crypto Trading and Bitcoin Prices: Evidence From a New Database of Retail Adoption," *SSRN Journal* 10266 (2023): 1–28, <https://doi.org/10.2139/ssrn.4357559>.
33. I. Madan, S. Saluja, and A. Zhao, "Automated Bitcoin "Trading via Machine Learning Algorithms"," accessed March 28, 2024, p. 5, <https://cs229.stanford.edu/proj2014/Isaac%20Madan,%20Shaurya%20Saluja,%20Aojia%20Zhao,Automated%20Bitcoin%20Trading%20via%20Machine%20Learning%20Algorithms.pdf>.
34. "Efficient AI Crypto Trading Bot for Optimal Results," accessed August 5, 2024, <https://wundertrading.com/en/ai-crypto-trading-bot>.

35. "Crypto Arbitrage Bot Development—Crypto Trading Bot Development Company," Antier Solutions, accessed August 5, 2024, <https://www.antiersolutions.com/crypto-arbitrage-bot-development/>.
36. R. A. Caulk, E. Törnquist, M. Voppichler, et al., "FreqAI: Generalizing Adaptive Modeling for Chaotic Time-Series Market Forecasts," *Journal of Open Source Software* 7, no. 80 (2022): 4864, <https://doi.org/10.21105/joss.04864>.
37. K. Au, "Crypto Arbitrage," Mar. 22, 2023, https://github.com/kelvinau/crypto-arbitrage/blob/2f8956fe37b62002985edfba84006f19490697de/engines/exchange_arbitrage.py.
38. "scikit-learn: Machine Learning in Python—scikit-learn 1.5.2 Documentation," accessed November 1, 2024, <https://scikit-learn.org/stable/>.
39. K. Košťál, D. Morháč, E. Matovič, et al., (2025). "Detecting and Predicting Changes in Crypto Markets." *2025 37th Conference of Open Innovations Association (FRUCT)*, 145–152. <https://doi.org/10.23919/fruct65909.2025.11008029>.
40. "1.1. Linear Model," scikit-learn, accessed November 24, 2024, https://scikit-learn/stable/modules/linear_model.html.
41. "LogisticRegression," scikit-learn, accessed November 24, 2024, https://scikit-learn/stable/modules/generated/sklearn.linear_model.LogisticRegression.html.
42. "1.11. Ensembles: Gradient Boosting, Random Forests, Bagging, Voting, Stacking," scikit-learn, accessed November 24, 2024, <https://scikit-learn/stable/modules/ensemble.html>.
43. "RandomForestClassifier," scikit-learn, accessed November 24, 2024, <https://scikit-learn/stable/modules/generated/sklearn.ensemble>.
44. "SVC, scikit-learn," accessed November 24, 2024, <https://scikit-learn/stable/modules/generated/sklearn.svm>.
45. "1.4. Support Vector Machines," scikit-learn, accessed November 24, 2024, <https://scikit-learn/stable/modules/svm.html>.
46. "MLPClassifier," scikit-learn, accessed November 24, 2024, https://scikit-learn/stable/modules/generated/sklearn.neural_network.MLPClassifier.html.
47. "1.17. Neural Network Models (Supervised)," scikit-learn, accessed November 24, 2024, https://scikit-learn/stable/modules/neural_networks_supervised.html.

Appendix A

Detailed Hyperparameter Tuning

As part of the adjustment of ML models to be precisely designed for the desired task, hyperparameter tuning was performed as described in Section 5.1. A more thorough description of the examined possibilities and variations is given in order to prove the suitability of the selected hyperparameters for the predictions of arbitrage occurrence. The individual tables show the considered hyperparameter with all tested values, together with their most optimal value marked in red.

Firstly, the hyperparameters for logistic regression are analyzed and summarized in Table A1. The penalty term refers to calculating the penalty for the loss function minimization during the training process. By L_1 , the absolute value of the residuals is taken into consideration in comparison to the square root of their values by the L_2 penalty. As experimenting with penalty term is performed and the solvers, as optimization algorithms for the weight adjustment in the logistic regression model, depend on this term, only the default value “lbfgs” is used. Further, multiple values of parameter C are tried out to modify the strength of the regularization. The smaller values correspond to stronger regularization. The total number of iterations is restricted to prevent overfitting of the model [40, 41].

After the iteration of the training process on all datasets for every cryptocurrency pair and every time interval, the most suitable values of hyperparameters were chosen, namely:

1. L_2 penalty term for lbfgs solver,
2. C value of 10 for stronger regularization,
3. the maximum number of iterations is 500, higher values lead to overfitting.

Afterward, the hyperparameters of random forest are examined and summed up in Table A2. Here, the number of decision trees is determined by the parameter “n_estimators”. The process in the decision tree is moderated by the criterion that specifies individual decisions by measuring the quality of every proposed split. Here, the Gini impurity or entropy implemented via Shannon information gain is tried out. Aiming to prevent overfitting, the minimum number of samples that have to be left in the node is set by “min_samples_split” and the maximum number of features accessed through the training process is limited to the square root or logarithmic value of their full amount via “max_features”. In order to prevent the usage of the whole dataset for the building process of every decision tree, bootstrapping is always enabled [42, 43].

TABLE A1 | Hyperparameters for logistic regression.

Logistic regression					
penalty	L_1	L_2			
C	0.01	0.1	1	10	100
max_iter	500	1000	1500		

TABLE A2 | Hyperparameters for random forest.

Random forest				
n_estimators	50	100	150	
criterion	gini	entropy		
min_samples_split	2	10	50	
max_features	sqrt	log2	None	
bootstrap	True			

The training process on all available datasets resulted in the following set of hyperparameters:

1. “n_estimators” sets the total number of decision trees to 150,
2. “gini” is chosen as the split criterion,
3. “min_samples_split” limits the minimum number of samples in a node to 10,
4. “max_features” sets the number of features used during the training to the square root of the total amount.

The hyperparameters for the SVM model are proposed and evaluated in Table A3. The strength of regularization is controlled via C parameters identical to those of logistic regression. Also, the maximum number of iterations is limited in order to prevent overfitting. Multiple types of kernels, namely, linear and polynomial, are examined to modify the decision boundary between individual classes. The key difference between these options lies in the number of spatial dimensions to which the dataset features are expanded [44, 45].

The training process on all cryptocurrency pairs and all time intervals led to the selection of the combination of hyperparameters as stated below:

1. C parameter is set to 10 for stronger regularization,
2. the maximum number of iterations is limited to 1500,
3. the linear kernel type is used.

Lastly, the simplest model of neural network called MLP is trained, and the selected hyperparameters are highlighted in Table A4. Experimenting with typical parameters of neural network training is performed. Multiple values of activation functions to activate the neurons in hidden layers are tried out, namely, sigmoid, hyperbolic tan function, and ReLU. Further, multiple solvers are used, specifically SGD and adam, to control the weight optimization process. The initial value of the learning rate is adjusted as well. Similarly to previous models, the maximum number of iterations is limited due to overfitting prevention. The size of the network is set to the default value of two hidden layers with five neurons each to keep the network small and keep the focus of the paper on a simpler ML model [46, 47].

The training process resulted in the selection of hyperparameters that are the most appropriate for the provided datasets and is summarized below:

1. ReLU as activation function is used,
2. the solver is set to SGD,

TABLE A3 | Hyperparameters for support vector machine.

Support vector machine					
C	0.01	0.1	1	10	100
max_iter	500	1000	1500		
kernel	linear	poly			

TABLE A4 | Hyperparameters for multilayer perceptron.

Multilayer perceptron			
activation	logistic	relu	tanh
solver	sgd	adam	
learning_rate_init	0.001	0.01	0.1
max_iter	100	150	200
hidden_layer_sizes	(5, 5)		

3. 0.01 is used as the initial value of the learning rate,
4. the maximum number of iterations is limited to 200.

This hyperparameter selection is almost identical to the previous training phase on less volatile price data. The only difference lies in the maximum number of iterations for MLP that increased from 100 to 200. This suggests that MLP is better suited to more volatile periods, where more iterations can be run without overfitting the model. The rest of the hyperparameters were not affected by the change in the volatility of the cryptocurrency market.

Appendix B

Evaluation Metrics of Machine Learning Models

As more cryptocurrency pairs and more time intervals create multiple separate datasets, the results obtained from the training process of all the models are extensive. This appendix provides a more thorough evaluation of the models' abilities and justifies the recorded results. The following tables summarize the evaluation metrics for all the datasets covering both cryptocurrency pairs and all time intervals for each ML model individually. The values of precision, AUC score, F1 score, and accuracy are recorded for every trained model. The most suitable time interval for every traded pair, based primarily on the level of precision, is marked in red. The tables are an extended version of Table 4 described in Section 5.1.

Logistic regression was trained as the simplest model, and the obtained results are summarized in Table B1. Apparently, the evaluation of BTCUSDT is not satisfactory, especially due to the level of precision, which is considered to be the most valuable metric for the task of arbitrage predictability. These low values can be caused by the simplicity of the logistic regression model, which is not able to follow the rapidly changing trend of BTCUSDT price development. The model's outputs improve significantly with datasets covering the ETHUSDT pair, which is less volatile, and therefore, the model's predictability power is higher. The most accurate predictions are made with a 5-min time interval, and this ability is rapidly lowering as the time interval increases, highlighting the necessity for datasets with frequently measured values.

Secondly, the random forest model is trained identically to the process with logistic regression. The results of all runs and corresponding values of metrics are stored in Table B2. The results and their justification are roughly the same, with minor differences. The random forest is able to follow the changing trend of BTCUSDT more precisely, but interestingly, only with a 1-min time interval. This supports the claim that only very narrow time intervals are appropriate for rapidly moving cryptocurrencies. The metrics for ETHUSDT provide almost consistent results and justification, as by the logistic regression; only, similarly to BTCUSDT, the 1-min time interval can be followed more precisely.

Afterward, the SVM model was trained, and the corresponding values of metrics are stored in Table B3. Generally, the metrics show that this model is unsuitable for the studied task. The results of all observed metrics are significantly the lowest so far. Although similar to the previous models, the datasets covering ETHUSDT provide better results, supporting the claim that arbitrage opportunities for less volatile cryptocurrency pairs can be predicted with higher confidence. Surprisingly, the best results for the ETHUSDT pair are reached with a 15-min time interval, which does not align with the findings so far. However, the metrics still provide poor values, so the inappropriateness of the model for the whole task can justify this discrepancy.

Lastly, the simplest structure of NN represented via MLP was trained. The complete evaluation of the model on all available datasets is summarized in Table B4. As the prior focus of the evaluation is on the precision metric, this model provides the best results for both cryptocurrency pairs. The values of other metrics are, in some cases, also higher than with the previously trained models or provide values without any statistically significant deviation. For both pairs, the 5-min time interval between individual data records resulted in the highest confidence in arbitrage predictability. Identically to all the previous models, the pair of ETHUSDT can be predicted with higher precision, which was already justified by the impact of lower volatility.

In summary, the best results were obtained when training the MLP model in a 5-min interval between individual data records. All metrics, except for accuracy, show significantly higher values for the ETHUSDT pair, assuming that arbitrage opportunities in a less volatile environment can be predicted with higher confidence and consequently lead to higher profits when utilized by an arbitrage bot.

TABLE B1 | Results of logistic regression training.

Cryptocurrency	Time interval	Precision	AUC score	F1 score	Accuracy
BTCUSDT	1 min	0.0547	0.942	0.1032	0.9727
BTCUSDT	5 min	0.0783	0.8537	0.1413	0.9847
BTCUSDT	15 min	0.0909	0.6205	0.1333	0.9884
ETHUSDT	1 min	0.6903	0.9841	0.813	0.9799
ETHUSDT	5 min	0.7943	0.9716	0.8676	0.9859
ETHUSDT	15 min	0.6337	0.9361	0.742	0.9736

TABLE B2 | Results of random forest training.

Cryptocurrency	Time interval	Precision	AUC score	F1 score	Accuracy
BTCUSDT	1 min	0.2113	0.914	0.3371	0.9943
BTCUSDT	5 min	0.0305	0.7323	0.0578	0.9684
BTCUSDT	15 min	0.0284	0.7195	0.0583	0.9374
ETHUSDT	1 min	0.7846	0.9866	0.8737	0.9874
ETHUSDT	5 min	0.8017	0.9739	0.8736	0.9866
ETHUSDT	15 min	0.7086	0.9257	0.7799	0.9792

TABLE B3 | Results of support vector machine training.

Cryptocurrency	Time interval	Precision	AUC score	F1 score	Accuracy
BTCUSDT	1 min	0.0748	0.5598	0.0928	0.9959
BTCUSDT	5 min	0.0032	0.5375	0.0062	0.907
BTCUSDT	15 min	0.0029	0.4311	0.0057	0.28
ETHUSDT	1 min	0.3946	0.7669	0.4679	0.9422
ETHUSDT	5 min	0.2988	0.82	0.4235	0.9047
ETHUSDT	15 min	0.5614	0.9321	0.69	0.9659

TABLE B4 | Results of multilayer perceptron training.

Cryptocurrency	Time interval	Precision	AUC score	F1 score	Accuracy
BTCUSDT	1 min	0.0901	0.9585	0.1644	0.9836
BTCUSDT	5 min	0.3333	0.6384	0.303	0.9978
BTCUSDT	15 min	0.0387	0.7278	0.0719	0.954
ETHUSDT	1 min	0.7463	0.986	0.8501	0.9846
ETHUSDT	5 min	0.8202	0.9676	0.8785	0.9874
ETHUSDT	15 min	0.6531	0.937	0.7552	0.9754

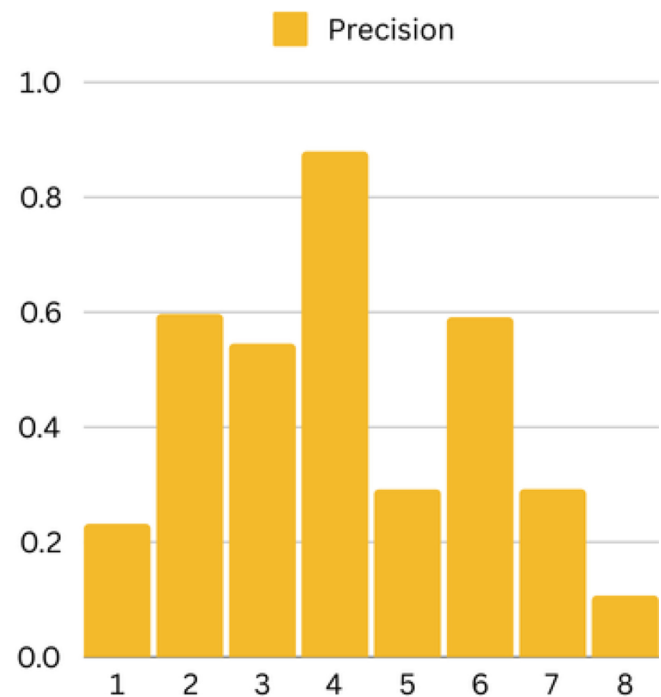
Appendix C

Normality Distribution of the Data for Hypotheses Testing

For the purpose of hypothesis testing, the normality of the data needed to be verified. The normal distribution was verified using the Shapiro–Wilk test, and the results represented via p -value are already present in Table 6. In order to prove the normal distribution graphically as well, the input data for the Shapiro–Wilk test are visualized. These input data correspond to the values of precision metrics during the training process. These values are divided based on the datasets' time interval or cryptocurrency pair. The division by the time interval causes three sets of precision values that serve to test the hypothesis for a time interval, aiming to verify the impact of the time interval on the model's prediction ability. The second division by the cryptocurrency pair creates two sets, one including precision values for datasets with BTCUSDT prices representing the more volatile pair, and the second one covers the precision values for ETHUSDT as the less volatile pair. The hypothesis then uses this division for the volatility of a cryptocurrency pair, which aims to prove the impact of the volatility.

For the purpose of the hypothesis for a time interval, Figure C1 visualizes the precision values of the training process on all datasets covering a 1-min time interval. Based on the Shapiro–Wilk test, the obtained p -value is 0.0733, which suggests that the data are still able to follow the normal distribution with a confidence level of 95%. The visualization supports the claim that the data still slightly follows the normal distribution. Secondly, Figure C2 aims to show the same fact for 5-min time interval. The Shapiro–Wilk testing provided a p -value of 0.0561 that is even closer to rejecting the satisfaction of normal distribution, but the data are still able to follow its trend. This claim is supported by the visualization as well. Lastly, the 15-min time interval is visualized in Figure C3. Here, the p -value of 0.0461 returned by the Shapiro–Wilk test rejects the assumption of normal distribution, and the visualization of the data provides the same claim.

The second hypothesis, dedicated to the impact of cryptocurrency volatility, requires two sets of data. Figure C4 visualizes the values of precision metric for all datasets covering the BTCUSDT pair. Based on the Shapiro–Wilk test with a p -value of 0.0469, the assumption of the

**FIGURE C1** | Precision for 1-min time interval.

normal distribution is not met, which is supported by the visualization as well. Almost identically, Figure C5 shows the values of precision for the ETHUSDT pair that show the absence of normal distribution, too. This is supported by the p -value of 0.0046 obtained by the Shapiro–Wilk test.

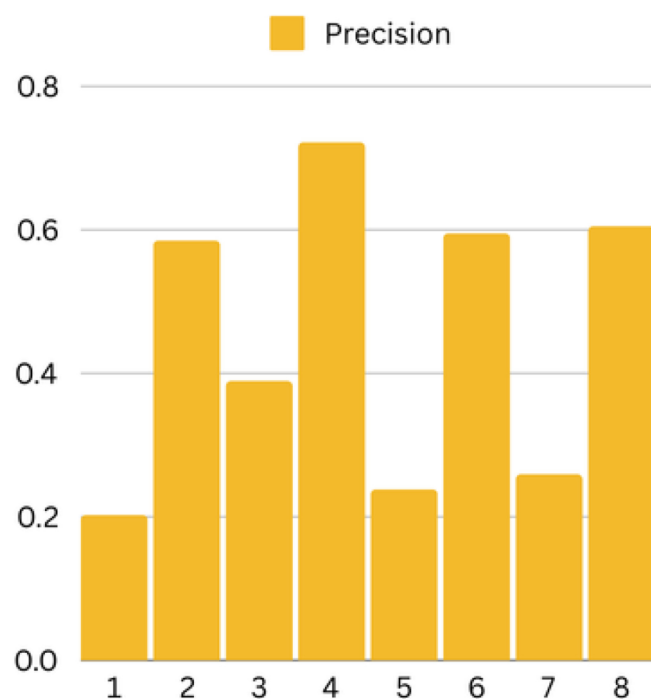


FIGURE C2 | Precision for 5-min time interval.

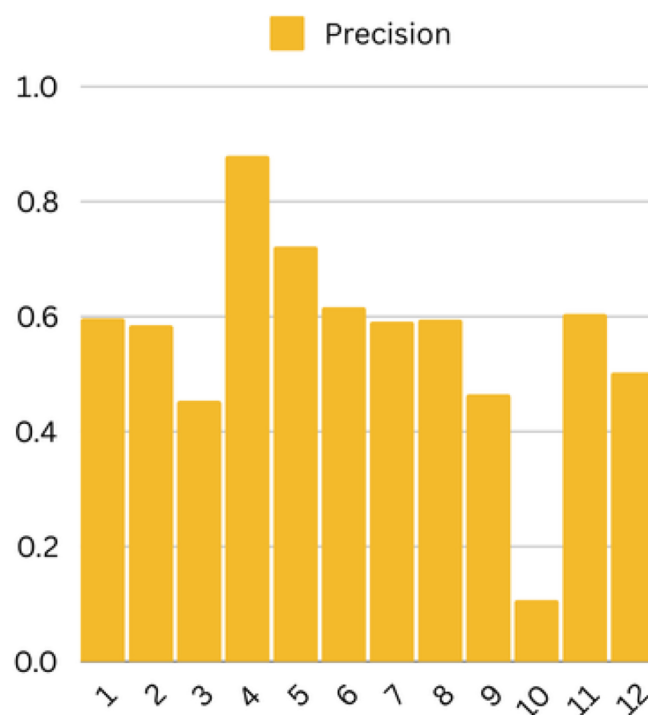


FIGURE C4 | Precision for less volatile cryptocurrency pair.

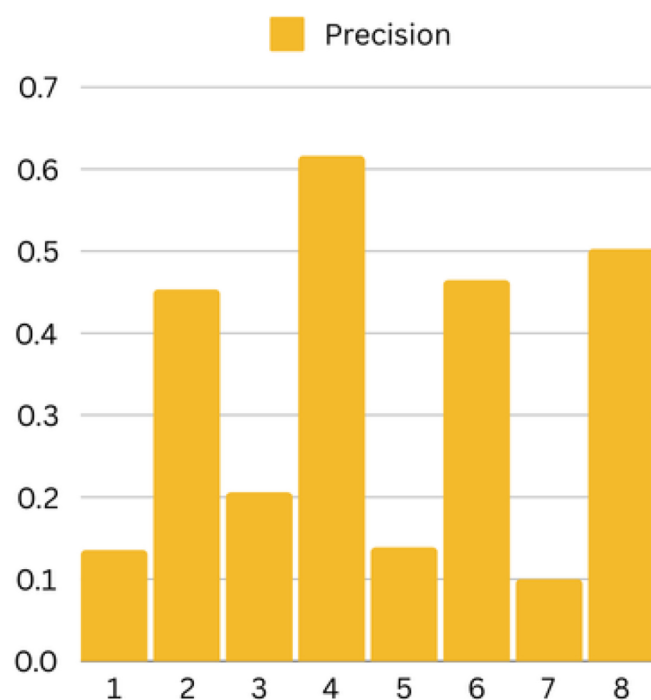


FIGURE C3 | Precision for 15-min time interval.

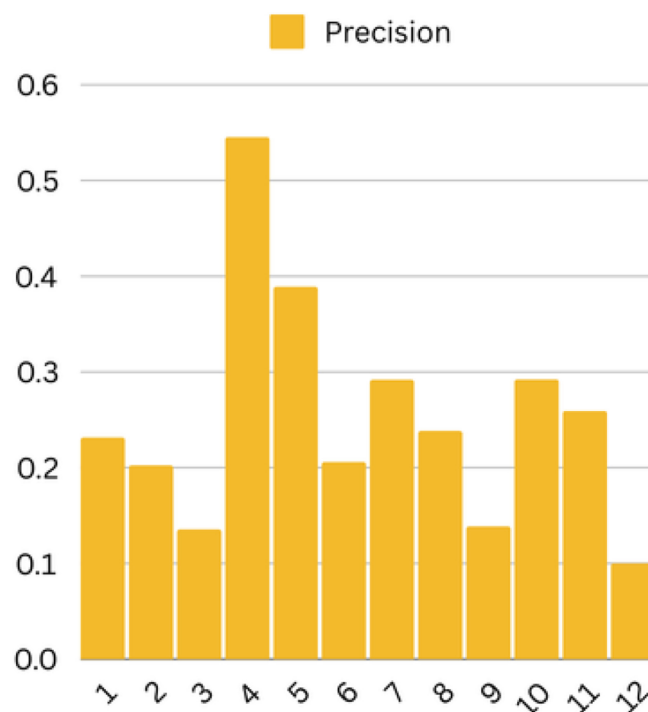


FIGURE C5 | Precision for more volatile cryptocurrency pair.

Appendix D

Structure of the Dataset

For the purpose of training ML models, a reasonable amount of data has to be collected and structured adequately into a dataset. The paper collects price data from two cryptocurrency exchanges, namely, Binance and Bybit, and for two cryptocurrency pairs, BTCUSDT and ETHUSDT, for the past half year. The frequency of the data collection results in the emergence of multiple datasets for 1-, 5-, and 15-min time intervals. Further, the preprocessing step of the models' training results

in additional features, such as the change of the price value and the possible indication of arbitrage occurrence.

The dataset is constructed separately for every cryptocurrency pair and every time interval, resulting in six separate datasets as summarized in Table D1. Every dataset consists of concatenated price values from both exchanges and the calculated probability of profitable arbitrage. After the preprocessing step, the dataset's structure is described in Table D2 with additional information about the individual features and their origin. The datasets in the defined format are being further used to train the selected models.

TABLE D1 | Individual datasets used for ML training.

Dataset name	Cryptocurrency pair	Time interval	Size (in megabyte [MB])
BTCUSDT_1m.csv	BTCUSDT	1 min	44.1 MB
BTCUSDT_5m.csv	BTCUSDT	5 min	8.8 MB
BTCUSDT_15m.csv	BTCUSDT	15 min	2.9 MB
ETHUSDT_1m.csv	ETHUSDT	1 min	43.3 MB
ETHUSDT_5m.csv	ETHUSDT	5 min	8.6 MB
ETHUSDT_15m.csv	ETHUSDT	15 min	2.8 MB

TABLE D2 | Structure of the datasets after the preprocessing phase.

Column	Description
date	The date that corresponds to the price data in format YYYY-MM-DD HH:MM:SS
dateTime	The datetime that corresponds to the price data
open_Binance	The opening price on Binance for the selected time interval
high_Binance	The highest price on Binance in the selected time interval
low_Binance	The lowest price on Binance in the selected time interval
close_Binance	The closing price on Binance for the selected time interval
volume_Binance	The traded volume on Binance in the selected time interval
open_Bybit	The opening price on Bybit for the selected time interval
high_Bybit	The highest price on Bybit in the selected time interval
low_Bybit	The lowest price on Bybit in the selected time interval
close_Bybit	The closing price on Bybit for the selected time interval
volume_Bybit	The traded volume on Bybit in the selected time interval
change_Binance	The percentage change between the current and previous open price on Binance
change_Bybit	The percentage change between the current and previous open price on Bybit
arbitrage	The possibility of profitable arbitrage occurring, value of 0 represents no profitable arbitrage and value of 1 suggests existence of arbitrage opportunity